

1. Dataset Overview

You have been provided with a dataset named `customer_data.csv` containing 100,000 rows. This dataset includes various numerical and categorical features related to customer profiles, income, and purchasing behavior.

Columns in the Dataset:

- **customer_id**: Unique identifier for each customer.
- **age**: Customer age (continuous numerical).
- **gender**: Customer gender (Male, Female, Other).
- **country**: Customer location (USA, UK, Canada, Australia, India).
- **education_level**: Education (High School, Bachelor, Master, PhD).
- **subscription_type**: Plan type (Free, Basic, Premium).
- **income**: Annual income in USD (highly skewed numerical).
- **credit_score**: Credit score ranging roughly from 300 to 850 (numerical).
- **purchase_amount**: Total amount spent (numerical).

2. Lab Tasks

Load the dataset using pandas and perform the following data transformation tasks. Create new columns for the transformed data so you can compare them with the originals.

1. Task 1: Discretization (Binning)

The age column contains continuous values. Discretize the age into three equal-width bins or logical buckets: *Young* (18-35), *Adult* (36-55), and *Senior* (56+). Save this as `age_binned`.

2. Task 2: Label Encoding

The gender and subscription_type columns contain categorical text data. Use Label Encoding to convert these into integer values. Save them as `gender_le` and `subscription_le`.

3. Task 3: One-Hot Encoding

The country column represents nominal categorical data without any inherent order. Apply One-Hot Encoding to create dummy variables for

each country. Ensure you handle the dummy variable trap if using linear models later (drop first).

4. **Task 4: Ordinal Encoding**

The education_level column has an inherent order (High School < Bachelor < Master < PhD). Use Ordinal Encoding to map these correctly to integers (0, 1, 2, 3) representing their rank. Save as education_ordinal.

5. **Task 5: Z-Score Standardization**

The credit_score column operates on a specific scale. Standardize it using Z-Score Standardization so that it has a mean of 0 and a standard deviation of 1. Save as credit_score_z.

6. **Task 6: Min-Max Normalization**

Normalize the purchase_amount column so that all values fall strictly between 0 and 1. Save as purchase_amount_minmax.

7. **Task 7: Mean Normalization**

Apply Mean Normalization to the income column. The formula is: $(x - \text{mean}) / (\text{max} - \text{min})$. Save as income_mean_norm.

8. **Task 8: Log Scaling**

The income column is heavily right-skewed. Apply Log1p Scaling (Logarithmic transformation) to compress the long tail and make the distribution more normal. Save as income_log.

9. **Task 9: Decimal Scaling**

Apply Decimal Scaling to the purchase_amount column. Determine the maximum absolute value in the column, find the smallest integer j such that the maximum absolute value divided by 10^j is less than 1, and divide all values by 10^j . Save as purchase_amount_decimal.